

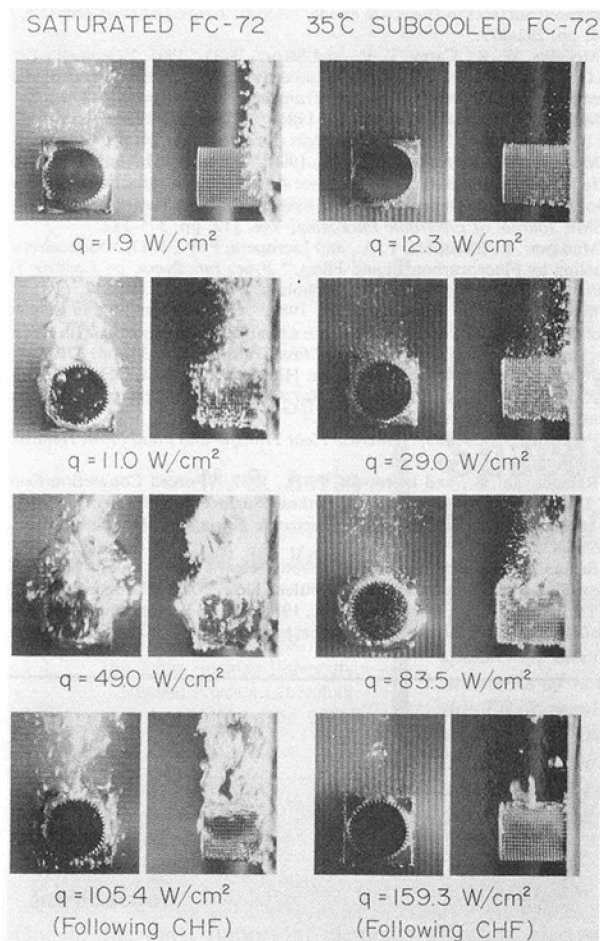


Fig. 14 Comparison of the development of boiling pattern on the microstud-enhanced stud in saturated and subcooled FC-72

liquid interface of the vapor layer surrounding the enhanced surface during film boiling also reduced the amount of vapor considerably, and vapor was released from the stud in relatively small bubbles which diminished in size following departure into the subcooled liquid.

Very important practical conclusions may be drawn from these subcooled boiling experiments. First, subcooling significantly enhances boiling performance and increases CHF. Subcooling may be essential in controlling bubble boundary layer growth on high power multichip modules.

**4.3 CHF Results.** Figure 15 shows a compilation of CHF data normalized with respect to CHF for a vertical flat surface (20.3 W/cm<sup>2</sup>) for all the surfaces examined in the present study. Each surface is represented by its value of  $A_r^+$ , the ratio of total surface area to planform base area provided in Table 4.

The dimensionless CHF shown in Fig. 15 is a measure of "CHF fin efficiency" since

$$q_m^+ = \frac{q_m A_{\text{chip}}}{q_{m, \text{flat}} A_f} = \frac{q_m}{q_{m, \text{flat}}} (A_r^+)^{-1} \quad (12)$$

The normalized CHF value  $q_m^+ = 1$  corresponds to surfaces whose CHF enhancement over a flat surface is balanced by the increased conduction resistance within the extended surface. The conduction resistance raises the base temperature, promoting the formation of a localized vapor blanket in that region. Only the flat microstud surface effectively delayed CHF since it increased surface area considerably with only a mild increase in fin volume compared with the pin fin and enhanced stud surfaces.

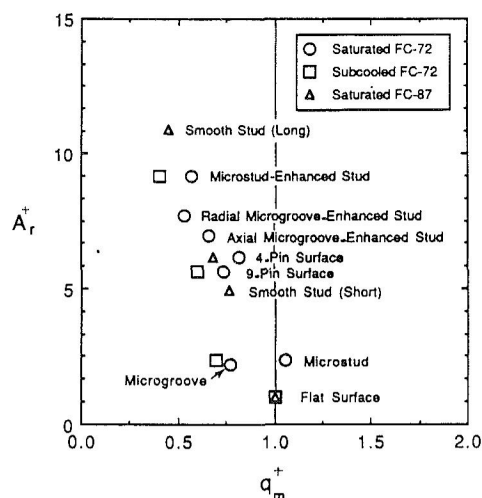


Fig. 15 CHF fin effectiveness for all the enhanced surfaces of the present study

## Summary

The present study involved an examination of several surface geometries for the purpose of enhancing heat transfer from high power electronic chips. Specific results from the study are as follows:

1. The heat transfer performances of horizontal boiling cylindrical fins can be successfully predicted using two-dimensional axisymmetric heat diffusion models. The assumption that localized boiling heat transfer from an extended surface is similar to that from an isothermal surface at the same temperature, regardless of surface orientation, was proven effective in predicting boiling performance, except for low post-incipience heat fluxes.

2. The elemental analysis based on extending single fin numerical results to fin arrays was successful in predicting boiling curves for 4-pin and 9-pin enhanced surfaces. This approach constitutes a practical tool in the design of enhanced surfaces for multichip modules.

3. Combining different levels of enhancement is an effective means of improving chip heat dissipation rate. An optimum enhanced surface was constructed from a single extended cylindrical stud with microstructures machined into the stud perimeter, followed by blasting the surface with an air/water/silica slurry.

4. Subcooling is very effective in enhancing boiling performance and increasing CHF. Subcooling may be necessary for controlling bubble boundary layer growth on high power multichip modules.

5. The combined use of extended surfaces and subcooling is a viable alternative for cooling very high power electronic chips. Microstructured extended surfaces resulted in chip heat fluxes (based on planform area) of 105.4 and 159.3 W/cm<sup>2</sup> in saturated and 35°C subcooled FC-72, respectively.

## Acknowledgment

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## References

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